Reducing late deliveries: where to start

To: Operations Manager

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Data: Olist marketplace, 96,470 delivered orders bought from September 2016 to August 2018

Late orders get much worse reviews. 6.8% of delivered orders arrived after the estimated date. These late orders averaged 2.27 stars, while on-time orders averaged 4.29. 62.4% of late orders got a 1- or 2-star review, but only 9.3% of on-time orders did. Even being 1 to 3 days late lowers the average to 3.29. My recommendations are below. The first two are about where most late orders come from, and the third is about high shipping costs on small orders.

1. Improve carriers in Rio de Janeiro and the Northeast, and plan for busy months

Rio de Janeiro has a 12.1% late rate on 12,350 orders. In São Paulo it is 4.5%. Rio alone makes up 22.9% of all late orders. Northeast states have even higher late rates on fewer orders, like 21.4% in Alagoas and 17.4% in Maranhão. 72.2% of late orders were handed to the carrier on time, so most of the delay happens during shipping. Late rates also went up in November 2017, the busiest month (7,288 orders, 12.4% late), and in February and March 2018 (14.1% and 19.0%).

What to do: track the on-time rate of carriers in Rio de Janeiro and the Northeast every week, by state, and update the delivery estimates on routes that keep missing them. For the busy season, line up extra carrier capacity before November.

2. Make sellers ship on time

9.0% of orders reached the carrier after the seller's deadline. These orders were late 20.8% of the time. When the seller shipped on time, the late rate was 5.4%. Orders that sellers shipped late make up 27.8% of all late orders. Many different sellers miss the deadline. The 31 sellers with at least 50 orders and at least double the average late rate make up only 8.6% of late seller-orders.

What to do: send sellers automatic reminders before the deadline and add on-time shipping to the seller scorecard. Sellers who keep missing it should get bigger penalties.

3. Test a minimum order amount on orders under 50 BRL

Orders with less than 50 BRL of items are 29.8% of delivered orders. On these orders, freight is 32.6% of what the customer pays. Across all orders it is 14.3%.

What to do: test free or cheaper shipping above a minimum order amount, and suggest add-on items at checkout. Compare order value and conversion with a control group.

Limitations

The data is a public sample from 2016 to 2018, so it may not match operations today. "Late" means delivered after the estimate shown to the customer, and I don't know how the estimates were set. Freight is what customers were charged for shipping, which isn't the same as what it actually cost. Revenue here means the value of the products, and Olist's commission is a separate thing. Late deliveries and low reviews go together, but this analysis does not prove that one causes the other. Orders still on the way when the data was pulled are left out, so the late rates for July and August 2018 are probably too low.